

# Fully Digital Feedforward Background Calibration of Clock Skews for Sub-Sampling TIADCs Using the Polyphase Decomposition

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**Abstract**—This paper presents a low-power fully digital clock skew feedforward background calibration technique in sub-sampling Time-Interleaved Analog-to-Digital Converters (TIADCs). Both estimation and correction algorithms share the common derivative filter, which makes them possible to reduce the chip area. Furthermore, these algorithms use the polyphase filtering technique and do not use adaptive digital synthesis filters. Thus, the proposed calibration can be implemented on a moderate hardware cost with low power dissipation. The adopted feedforward technology eliminates the stability issues encountered with the adaptive technique. The Hardware Description Language (HDL) design of the proposed calibration is synthesized using a 28nm FD-SOI process for a 60dB SNR TIADC clocked at 2.7GHz. The calibration is designed for both baseband and sub-sampling TIADC applications. For sub-sampling TIADCs with the input at the first four Nyquist bands, the synthesized calibration system occupies  $0.04\text{mm}^2$  of area and dissipates a total power of 33.2mW. For the baseband TIADC applications, it occupies  $0.02\text{mm}^2$  and consumes 15.5mW.

**Index Terms**—All-digital feedforward calibration, FPGA/ASIC implementation, polyphase filtering, subsampling and undersampling TIADCs.

## I. INTRODUCTION

IN MODERN communication systems such as broadband satellite receivers, cable TV tuners and Software-Defined Radios, Analog to Digital Converters (ADCs) play an essential role. Such ADCs require very high sampling rates with high resolution and low power dissipation. A Time-interleaved ADC (TIADC) which is formed by combining several slow but accurate ADCs in parallel, is a promising solution to achieve these goals [1], [2]. Unfortunately, TIADC performance suffers from channel mismatches including offset, gain and timing mismatches due to process, voltage, and temperature variations. These channel mismatches produce frequency aliases on the output signal and hence, significantly degrade the Signal-to-Noise and Distortion Ratio (SNDR)

and Spurious-Free Dynamic Range (SFDR) performance of time-interleaved ADCs. Among these aforementioned errors, sample-time errors are the most critical as the timing mismatch errors increase with the input frequency and overshadow the effect of other mismatches for broadband inputs [3]. For these reasons, this framework focuses on tackling the timing mismatch problem.

The timing mismatches in TIADCs can be effectively corrected by using analog and/or mixed signal calibration techniques [1], [4]–[9]. Mixed-signal calibration techniques mitigate the clock skew effect by adjusting the variable-delay line in clock buffers. They exhibit good performance but the analog correction schemes limit the overall ADC resolution due to variations in process, supply voltage and temperature as well as thermal noise. Moreover, they require an additional analog circuit with additional development time and are not portable between CMOS technology nodes.

Thanks to technology node shrinking, fully digital techniques seem to be a more promising solution to overcome the above issues of the analog and mixed-signal calibration. They can be developed faster, they make use of the advantages of CMOS technology scaling, and they are easier to port to the next technology generation. Most calibration approaches [10]–[17], are derived assuming an input signal band-limited to the Nyquist frequency, i.e., the input located in the first *Nyquist Band (NB)*, also commonly known as *Nyquist Zone (NZ)*. However, these techniques can not be directly applied in the undersampling (or subsampling) TIADCs which samples the band-limited signals in the higher NB. Undersampling is an interesting solution in next-generation direct-sampling receivers such as sub-sampling receivers, software defined radios, and broadband satellite receivers [18], [19]. A few all-digital calibration techniques coping with mismatches in undersampling TIADCs have been proposed in [19]–[22]. Nevertheless, they require either an additional channel [21] or a pilot input signal [22] for calibration. The work in [20] is performed with the assumption of narrow-band signals and applied for only two channels. For very high speed applications, the calibration in [19] used the polyphase filtering technique in order to enhance the working frequency of Digital Signal Processing (DSP) units. However, the use of two identical derivative filters (one in correction and one in estimation) is not optimal in terms of power consumption. In order to reduce the chip area, sharing one ideal differentiator

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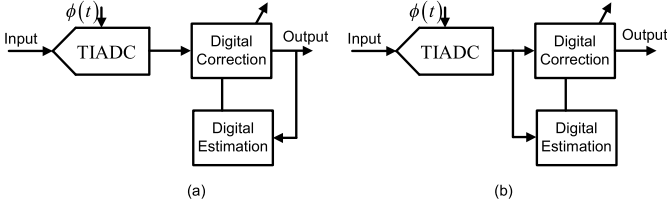


Fig. 1. All-digital calibration of timing mismatch in TIADCs: (a) all-digital feedback calibration and (b) all-digital feedforward calibration.

filter between correction and estimation schemes is proposed in this framework.

Regarding the all-digital calibration implementation, calibration schemes are performed either with the aid of a feedback loop as shown in Fig. 1(a) or in a feedforward manner as shown in Fig. 1(b) [23]. Therefore, the former is defined as *all-digital feedback* calibration and the latter as *all-digital feedforward* calibration. Most prior arts [11]–[17], [19]–[22] used adaptive feedback techniques which suffer from potential stability issues [10], [24]. In order to overcome these issues and to lower the power dissipation, the calibration algorithms in [10], [25] are performed in the feedforward manner. This approach indeed calculates the cross-correlations among the adjacent sub-ADCs and the cross-correlations between the sub-ADC output samples and their corresponding derivative samples. Using the orthogonal property of the input signal and its derivative, these cross-correlations enable a direct estimation of clock skews. The above pairwise cross-correlations are achieved by using the overall TIADC output and a FIR filter whose coefficients are  $-1, 0, +1$ . The sub-ADC derivative is obtained by feeding the TIADC gain-corrected samples into a FIR baseband derivative filter [10]. With such an implementation, the FIR filters are required to run at full sampling rate, which is very challenging and power hungry for very high speed ADC designs. Moreover, this technique does not work with input at any NZ.

Inspired by the all-digital feedforward calibration scheme of timing mismatch proposed in [10], [25], we present its theoretical analysis for sub-sampling TIADCs at any Nyquist Band in this paper. In addition, the complexity of the estimation scheme is reduced by using one cross-product between the sub-ADC output and the output derivative of the adjacent sub-ADC instead of using two cross-products in the computation of the derivative of the input autocorrelation. Furthermore, we leverage our previous work [19] by using polyphase implementation for both estimation and correction algorithms to reduce the power. Finally, the digital feedforward background calibration system is synthesized in a 28nm FD-SOI process.

In detail, a linear equation system of the clock skews is derived from the pairwise cross-correlations of sub-ADC outputs which are obtained from the direct sub-ADC output samples. The solutions of these established equations are the clock skew estimates. In addition, in both correction and estimation schemes, the derivatives of sub-ADC outputs are required and computed by using a polyphase technique. Note that the estimation and correction algorithms run at the sub-ADC rate, leading to the low power consumption. They also share the same derivative filter, i.e., the proposed algorithm saves a large

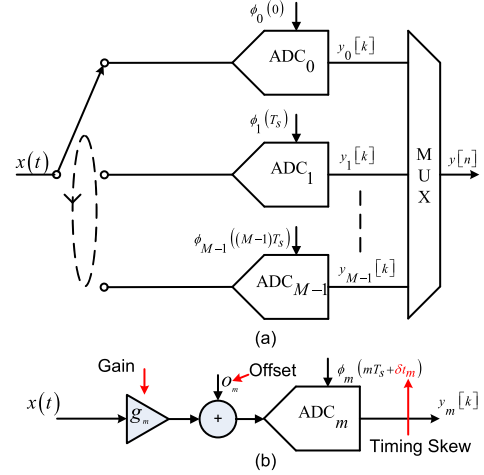


Fig. 2. (a) An ideal  $M$ -channel TIADC model; (b) the  $m^{th}$  nonideal sub-ADC channel model.

amount of hardware cost compared to our previous works proposed in [19], [26]. Furthermore, they do not use the adaptive filter banks. As a result, the proposed calibration alleviates the power consumption and chip die area. It is also employed for very high speed TIADC design (beyond several gigahertz).

The rest of the paper is organized as follows. Section II reviews time-interleaved ADCs, their limitations and timing skew induced errors. Section III presents the proposed feedforward calibration of the timing offset including the correction and estimation. In order to show the efficiency of the proposed calibration, Section IV analyzes simulation results, ASIC synthesis results and simulation results after processing the real data captured from the ADC chip. Conclusions are finally drawn in Section V.

## II. TIME-INTERLEAVED CONVERTERS

Fig. 2(a) shows a simplified block diagram architecture of a time-interleaved ADC. Its operation consists in using  $M$  time-interleaved sub-ADCs sampling the input signal at the same frequency  $\frac{f_s}{M}$ , but with equally phase-shifted clocks. The outputs of sub-ADCs are multiplexed to form a stream of the output data. Ideally, the overall ADC will have the same resolution as sub-ADCs, an equivalent sampling frequency  $f_s$ , and an equivalent sampling period  $T_s$ . However, channel mismatches among the constituent sub-ADCs including offset, gain and timing mismatches significantly degrade the linearity performance of TIADCs [3] as drawn in Fig. 2(b). Among the aforementioned mismatches, the timing mismatch also called the clock skew is the most critical error due to its frequency dependency. It results in a phase shift of the input samples which causes the arising of scaled replicas of input derivative at frequencies  $\pm f_{in} + k \frac{f_s}{M}$  where  $f_{in}$  is the input frequency and  $k$  is an integer.

In our calculation, for the sake of simplicity, we assume that there are no gain/offset mismatches.  $\delta t_i$  denotes the clock skew normalized with period  $T_s$  of the  $i^{th}$  channel ADC. Ignoring quantization effects, the digital output sequence  $y_m[k]$  of the  $m^{th}$  channel ADC is expressed by

$$y_m[k] = y_m(kT_s) = x(kMT_s + mT_s + t_0 + \delta t_m). \quad (1)$$

where  $t_0$  is the initial sampling phase.

Generally, the clock skews are small compared with the period  $T_s$ , typically less than 1%. Using the Taylor series approximation, the  $m^{\text{th}}$  channel sub-ADC output can be expressed as the sum of an ideal signal and an error term proportional to the timing offset and the signal derivative as [4], [10], [13]

$$y_m[k] \approx x_m[k] + \delta t_m x'_m[k], \quad (2)$$

where  $x'_m[k]$  is the derivative of the input signal at the nominal sampling time (or the derivative of the sub-ADC output in the case without timing skews). The timing skew induced error is a product of the derivative signal of sub-ADC output samples and its corresponding clock skews.

### III. PROPOSED CALIBRATION TECHNIQUE

The proposed calibration for the input at the first NZ consists of two main algorithms: correction and estimation presented hereafter. The proposed calibration extension to an input at any NZ is also described below.

#### A. Digital Correction

Once the estimate of  $\hat{\delta t}_m$  is known, it can be used in various ways to get the correct value of the input signal at the nominal sampling time. Some approaches use a digital synthesis filter bank in the correction as in [14], [20] that runs at full speed all the time and requires large amounts of computation resources. Therefore, these approaches are not very suitable for low power specifications. In this framework, the adopted approach is to subtract the error given by the second term of (2) from the conversion result. This would require a knowledge of the derivatives  $x'_m[k]$  of ideal sub-ADC outputs. In blind calibration, the input is unknown, i.e.,  $x'_m[k]$  is also unknown. Thus, the derivatives  $y'_m[k]$  of the distorted sub-ADC output samples are used for correction as done in [10], [11], [13], [24], [27]. Derivatives  $y'_m[k]$  are computed in the digital domain using polyphase filters of the digital differentiators proposed in [19], [26]. With the 25-tap FIR differentiator used in [26], the derivative span is extended up to 90% of the Nyquist zone before it rolls off.

#### B. Digital Estimation

Let the analog input of  $x(t)$  be a Wide-Sense Stationary (WSS) random process and band-limited to the Nyquist frequency, i.e., its mean  $E\{x(t)\}$  is constant and its autocorrelation  $R_x(\tau)$  depends only on the time shift  $\tau = t_1 - t_2$  [28]. These characteristics can be expressed by the following equations:

$$E\{x(t)\} = \eta = \text{constant}, \quad (3a)$$

$$R_x(t_1, t_2) = R_x(\tau) = E\{x(t + \tau)x(t)\}, \text{ for all } t. \quad (3b)$$

In the discrete-time domain, the cross-correlation  $R_{fg}[l]$  of two signals  $f[n]$  and  $g[n]$  is defined by [29], [30]

$$\begin{aligned} R_{fg}[l] &= E\{f[n+l]g[n]\} \\ &= \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N f[n+l]g[n], \end{aligned} \quad (4)$$

where index  $l$  is the (time) shift (or *lag*) parameter. The notation  $R_{fg}[l]$  is extensively used for writing convenience.

*Basic idea:* To estimate the clock skews of sub-ADCs, the cross-correlation of two adjacent channels of the TIADC must be computed. The difference between two cross-correlation functions produces a linear equation system whose variables are timing skews of the individual channel ADCs. The solution of this linear equation system is the timing skews of sub-ADCs. The estimation technique will be interpreted in more detail as follows.

1) *Linear Equation System With Clock Skew Variables:* From (4), (1) and (3b), the cross-correlation between two consecutive channel ADCs ( $\text{ADC}_m$  and  $\text{ADC}_{m-1}$  for  $1 \leq m \leq M-1$ ) is expressed by

$$\begin{aligned} R_{y_m y_{m-1}}[0] &= E\{y_m[k]y_{m-1}[k]\} \\ &= E\{x(kMT_s + mT_s + t_0 + \delta t_m) \\ &\quad \times x(kMT_s + mT_s - T_s + t_0 + \delta t_{m-1})\} \\ &= R_x(T_s + \delta t_m - \delta t_{m-1}). \end{aligned} \quad (5)$$

Analogously, the cross-correlation of two adjacent sequence outputs from sub-channel  $\text{ADC}_m$  and  $\text{ADC}_{m+1}$  for  $0 \leq m \leq M-2$  is written as

$$\begin{aligned} R_{y_m y_{m+1}}[0] &= E\{y_m[k]y_{m+1}[k]\} \\ &= E\{x(kMT_s + mT_s + t_0 + \delta t_m) \\ &\quad \times x(kMT_s + mT_s + T_s + t_0 + \delta t_{m+1})\} \\ &= R_x(T_s + \delta t_{m+1} - \delta t_m). \end{aligned} \quad (6)$$

In order to estimate sample-time errors, the timing error function is defined as the difference of pairwise cross-correlations of sub-ADC output samples and is expressed by

$$\Gamma_m = R_{y_m y_{m-1}}[0] - R_{y_m y_{m+1}}[0], \quad 1 \leq m \leq M-2. \quad (7)$$

By applying the Taylor series expansion up to the first derivative around a zero point, the autocorrelation function  $R_x(T_s + \delta t)$  as a function of the variables  $\delta t$  is expressed by

$$\begin{aligned} R_x(T_s + \delta t) &\approx R_x(T_s) + \delta t \times \left. \frac{dR_x(T_s + \delta t)}{d\delta t} \right|_{\delta t=0} \\ &\approx R_x(T_s) + \delta t \times R'_x(T_s), \end{aligned} \quad (8)$$

where  $R'_x(T_s)$  is the derivative of the autocorrelation of the input signal. In practice, the clock skews arising in time-interleaved ADCs are typically small compared to the sampling interval  $T_s$ . By applying Taylor series expansion to (5) and (6) as done in (8), the timing error function  $\Gamma_m$  in (7) can be simplified as follows.

$$\Gamma_m \approx R'_x(T_s)(2\delta t_m - \delta t_{m-1} - \delta t_{m+1}), \quad 1 \leq m \leq M-2. \quad (9)$$

Obviously, (9) is a linear equation system that has  $(M-2)$  equations and  $M$  variables being the timing skews of  $M$  channel sub-ADCs. In order to derive the linear equation system having the same number of linear equations as the number of variables, we introduce two additional timing error functions of  $\Gamma_0$  and  $\Gamma_{M-1}$  as follows.

$$\begin{aligned} \Gamma_0 &= R_{y_{M-1} y_0}[-1] - R_{y_0 y_1}[0]. \\ \Gamma_{M-1} &= R_{y_{M-1} y_{M-2}}[0] - R_{y_{M-1} y_0}[-1]. \end{aligned} \quad (10)$$

By substituting  $m = 0$  for  $m$  into (6), we have

$$R_{y_0 y_1}[0] = R_x(T_s + \delta t_1 - \delta t_0). \quad (11)$$

By replacing  $m$  with  $m = M - 1$  into (5), we have

$$R_{y_{M-1} y_{M-2}}[0] = R_x(T_s + \delta t_{M-1} - \delta t_{M-2}). \quad (12)$$

Analogously to (5) and (6), we completely compute

$$\begin{aligned} R_{y_{M-1} y_0}[-1] &= E\{y_{M-1}[k-1]y_0[k]\} \\ &= E\{x((k-1)MT_s + (M-1)T_s \\ &\quad + t_0 + \delta t_{M-1}) \\ &\quad \times x(kMT_s + 0 \times T_s + t_0 + \delta t_0)\} \\ &= R_x(T_s + \delta t_0 - \delta t_{M-1}). \end{aligned} \quad (13)$$

From (8), (10), (11), (12) and (13), we have

$$\begin{aligned} \Gamma_0 &\approx R'_x(T_s)(2\delta t_0 - \delta t_1 - \delta t_{M-1}). \\ \Gamma_{M-1} &\approx R'_x(T_s)(2\delta t_{M-1} - \delta t_{M-2} - \delta t_0). \end{aligned} \quad (14)$$

From (9) and (14), we have the following linear equation system with  $1 \leq m \leq M - 1$ :

$$\begin{cases} \Gamma_0 &\approx R'_x(T_s)(2\delta t_0 - \delta t_1 - \delta t_{M-1}) \\ \vdots &\vdots \\ \Gamma_m &\approx R'_x(T_s)(2\delta t_m - \delta t_{m-1} - \delta t_{m+1}) \\ \vdots &\vdots \\ \Gamma_{M-1} &\approx R'_x(T_s)(2\delta t_{M-1} - \delta t_{M-2} - \delta t_0). \end{cases} \quad (15)$$

In general, if the derivative of the input autocorrelation  $R'_x(T_s)$  is determined, solving the equation system (15) provides the sample-time error estimates of sub-ADCs.

2) *Estimates of the Input Autocorrelation Derivative:* From (3b), we have

$$R'_x(T_s) = \left. \frac{dR_x(\tau)}{d\tau} \right|_{\tau=T_s} = E\{x'(t+T_s)x(t)\}. \quad (16)$$

Without clock skews, the ideal sampling instant (or the ideal sampling edge) of the  $m^{th}$  sub ADC is at

$$t_{sm} = kMT_s + mT_s + t_0, 0 \leq m \leq M - 1. \quad (17)$$

By replacing  $t$  with  $t_{sm}$  into (16), we have

$$\begin{aligned} R'_x(T_s) &= E\left\{ \underbrace{x'(kMT_s + (m+1)T_s + t_0)}_{x'_{m+1}[k]} \right. \\ &\quad \left. \times \underbrace{x(kMT_s + mT_s + t_0)}_{x_m[k]} \right\} \\ &= E\{x'_{m+1}[k]x_m[k]\}, 0 \leq m \leq M - 1. \end{aligned} \quad (18)$$

where  $x_m[k]$  is the ideal output samples of the  $m^{th}$  channel ADC; and  $x'_{m+1}[k]$  is the derivative of the desired outputs of the  $(m+1)^{th}$  channel ADC. In a blind calibration mechanism, the input signal is not available, i.e.,  $x_m[k]$  and  $x'_{m+1}[k]$  are unknown. Thus, the output of the sub ADC <sub>$m$</sub>  and the derivative of the channel ADC <sub>$m+1$</sub>  are used to approximately compute the autocorrelation of the input signal. So the first

order derivative of the autocorrelation of the input signal is expressed as

$$\begin{aligned} R'_x(T_s) &\approx E\{y'_{m+1}[k]y_m[k]\} \\ &= R'_{y'_{m+1}y_m}[0], 0 \leq m \leq M - 1. \end{aligned} \quad (19)$$

In other words, the first order derivative of the autocorrelation of the input signal at the time  $T_s$  is computed by using one cross-correlation between the sub-ADC output and the derivative of its adjacent sub-ADC output. For the estimation technique in [10], [25], the derivative of the input autocorrelation is computed by an addition of two cross-correlations expressed in (19). Thus, our proposed estimation is less complex than the approach in [10], [25].

3) *Solution of the Linear Equation System:* The linear equation system (15) can be written in matrix notation as follows.

$$\frac{1}{R'_x(T_s)} \begin{bmatrix} \Gamma_0 \\ \Gamma_1 \\ \vdots \\ \Gamma_{M-1} \end{bmatrix} \approx \mathbf{H} \times \delta \mathbf{T}, \quad (20)$$

where  $\delta \mathbf{T}$  is a column vector whose elements are individual clock skews of sub-ADCs, and expressed by

$$\delta \mathbf{T} = [\delta t_0, \delta t_1, \dots, \delta t_{M-1}]^T \quad (21)$$

and

$$\mathbf{H} = \begin{bmatrix} 2 & -1 & 0 & \dots & 0 & -1 \\ -1 & 2 & -1 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ -1 & 0 & 0 & \dots & -1 & 2 \end{bmatrix}_{M \times M} \quad (22)$$

is a constant circulant  $M \times M$  matrix having the following properties.

- If the column index is equal to the row index, its corresponding coefficient is equal to 2.
- If  $ind_{col} = (ind_{row} \pm 1) \text{ modulo } M$ , its corresponding coefficients are equal to -1, where  $ind_{col}, ind_{row}$  are the indexes of the columns and the rows of matrix  $\mathbf{H}$ , respectively.
- The coefficients are equal to 0 for all other indexes of the rows and columns.
- $\mathbf{H}$  is a symmetric matrix, i.e.,  $\mathbf{H} = \mathbf{H}^T$ , where  $T$  notates the transpose operator.
- The circulant matrix  $\mathbf{H}$  is fully specified by its first row vector. The remaining rows are generated by the cyclic permutations of the first row. The final row is a vector in reverse order compared to the first row. Moreover, both the sum of all rows and the sum of all columns are zero. As a result, the rank of the circulant matrix  $\mathbf{H}$  is determined by [31] and is equal to  $(M - 1)$ .

Taking the sum of all equations of the above linear equation system results in

$$\frac{1}{R'_x(T_s)} (\Gamma_0 + \Gamma_1 + \dots + \Gamma_{M-1}) = 0, \quad (23)$$

which is not dependent on the variables of  $\delta t_m, 1 \leq m \leq M - 1$ . As a result, the linear equation (20) is simplified and



expressed in matrix notation by

$$\mathbf{A}\delta\mathbf{T} = \frac{1}{R'_x(T_s)}\mathbf{\Gamma}, \quad (24)$$

where

$$\mathbf{\Gamma} = [\Gamma_1, \dots, \Gamma_{M-1}]^T \quad (25)$$

and

$$\mathbf{A} = \begin{bmatrix} -1 & 2 & -1 & \dots & 0 & 0 \\ 0 & -1 & 2 & \dots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & 0 & 0 & \dots & -1 & 2 \end{bmatrix}_{(M-1) \times M} \quad (26)$$

is also a constant circulant  $(M-1)$ -by- $M$  matrix. The matrix  $\mathbf{A}$  is generated by deleting the first row of the constant matrix  $\mathbf{H}$ . From the properties of matrix  $\mathbf{H}$ , the circulant constant matrix  $\mathbf{A}$  is said to be *full row rank*, i.e.,  $\text{rank}(\mathbf{A}) = M-1$  [31]. Therefore, the  $(M-1)$ -by- $(M-1)$  square matrix  $\mathbf{A}\mathbf{A}^T$  is invertible, i.e., it has full rank,  $(M-1)$  [32].  $\mathbf{\Gamma}$  is a column vector of length of  $(M-1)$  whose elements are the defined timing error functions in Section III-B1. The  $M$ -by-1 vector  $\delta\mathbf{T}$  is a solution of the linear equation system (24) if  $\mathbf{A}\delta\mathbf{T} = \frac{1}{R'_x(T_s)}\mathbf{\Gamma}$  holds.

The system of linear equations (24) is *underdetermined* because there are more unknowns (variables  $\delta t_m, 0 \leq m \leq M-1$ ) than equations. This occurs once there are more columns than rows with linearly independent rows. In this case, the least-norm (or least square) solution of the underdetermined system of linear equations is given by the authors in [32]–[35]. The unique minimum norm solution (or the unique solution with minimal Euclidean norm) of the system can then be expressed as [32]–[35].

$$\delta\mathbf{T} \approx \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1} \frac{1}{R'_x(T_s)}\mathbf{\Gamma}. \quad (27)$$

$\mathbf{A}^\dagger = \mathbf{A}^T(\mathbf{A}\mathbf{A}^T)^{-1}$  is called the *pseudo-inverse* constant circulant matrix of full rank of  $M$ .  $\mathbf{A}^\dagger$  is also called a right inverse matrix of  $\mathbf{A}$ . The elements (or coefficients) of a minimum norm solution vector of the underdetermined equations are the clock skew estimates of sub-ADCs  $\delta t_m, 0 \leq m \leq M-1$ .

### C. Architecture of the Proposed Calibration

The overall architecture of the proposed calibration technique is shown in Fig. 3. It mainly consists of a *Derivative Polyphase Filter (DPF)* block proposed in [19], a *Cross-correlation Computation Unit (CCU)*, a *Gamma computation Unit ( $\Gamma$ U)* block, and a *Matrix Processing Unit (MPU)* block.

- **The Derivative Polyphase Filters (DPF):** From (19) and (2), the derivatives of the sub-ADC outputs are required to estimate the timing mismatch coefficients. The signal derivatives of sub-ADC outputs are obtained by a DPF block. The DPF block encompasses polyphase filter sub-systems  $\text{subPF}_i, 0 \leq i \leq M-1$  described in [19], [26]. The  $\text{subPF}_i$  is formed by delay elements  $z^{-1}$  and the type-1 polyphase filter components  $P_i(z)$  of the Nyquist derivative FIR filter. The impulse response of the  $i^{\text{th}}$  polyphase filter is given by

$$p_i[n] = h_d[nM+i], i = \{0, 1, \dots, M-1\}, \quad (28)$$

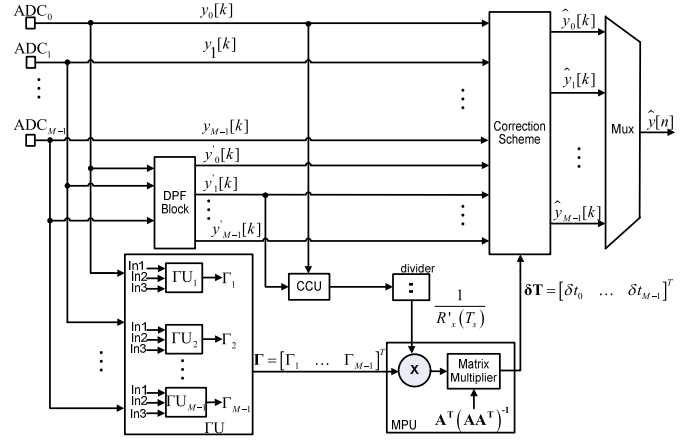


Fig. 3. The overall architecture of the proposed calibration.

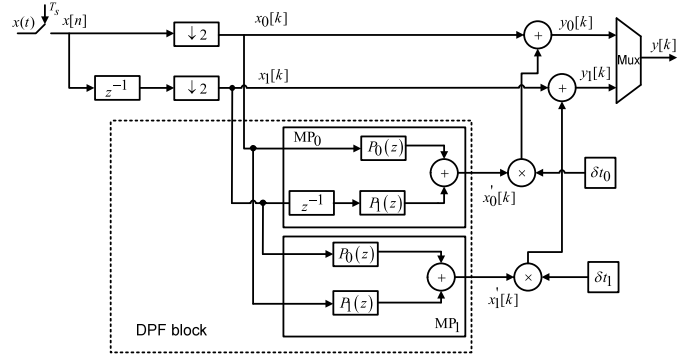


Fig. 4. Equivalent polyphase structure of the two-channel TIADCs [19], [26].

where  $h_d[n]$  is the impulse response of the ideal derivative filter and expressed as [29], [30]

$$h_d[n] = \begin{cases} \frac{\cos(n\pi)}{n} & (n \neq 0) \\ 0 & (n = 0). \end{cases} \quad (29)$$

The architecture of  $\text{subPF}_i$  is shown in Fig. 4 in the detail for an example of two-channel TIADCs. This architecture can be easily extended to  $M$ -channel TIADCs [19], [26].

- **Cross-correlation Computation Unit (CCU):** The *Cross-correlation Computation Unit (CCU)* is to compute the cross-correlation between two signals in (4). The limits in (4) can not generally be computed. In practice, given a lag  $l$ , the cross-correlation  $R_{fg}[l]$  is calculated by averaging the product of two signals over  $N$  samples as [29], [30]

$$\begin{aligned} R_{fg}[l] &= E\{f[n+l]g[n]\} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} f[n+l]g[n]. \end{aligned} \quad (30)$$

The cross-correlation in (30) is in fact a moving average of  $N$  cross-products of sub-ADC samples. The cross-product is computed by a digital multiplier. It is the input of the moving-average filter. In order to reduce the number of additions and memory requirements of the

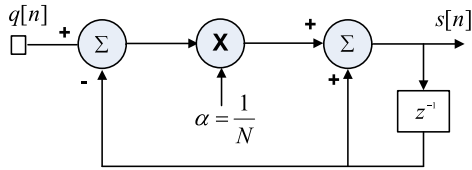


Fig. 5. The MMA filter.

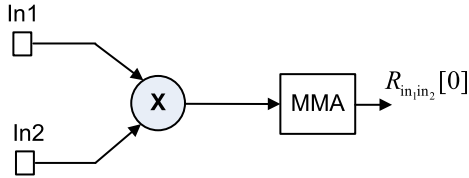


Fig. 6. Cross-correlation Computation Unit (CCU).

moving average direct computation of (30), a Modified Moving Average (MMA) filter [36] is used instead. It is defined by

$$\begin{aligned} s[n] &= \frac{N-1}{N}s[n-1] + \frac{1}{N}q[n] \\ &= (1-\alpha)s[n-1] + \alpha q[n], \end{aligned} \quad (31)$$

where  $s[n]$  and  $q[n]$  are its output and input, respectively. The coefficient  $\alpha = \frac{1}{N}$  represents a constant smoothing factor. Note that if  $N$  is large enough, (31) is an approximation of (30). The average output,  $s[n]$  is computed recursively based on the previous result  $s[n-1]$  and the output feedback term  $s[n-1]$ . The recursive equation given in (31) is widely used for moving average computation in DSP algorithms [36]. As a result, the MMA filter is drawn in Fig. 5. In hardware implementation, the period of  $N$  samples is selected to be  $2^k$  where  $k$  is a positive integer since multiplying the signal with  $\alpha$  is equivalent to a left arithmetic shift by  $k$  places, leading to outperform hardware multipliers.

Finally, the CCU block consists of the multiplier and the aforementioned MMA filter as shown in Fig. 6 for lag  $l = 0$ . Note that if the time shift  $l \neq 0$ , the delay is added to a corresponding input of multiplier in order to form the signal  $f[n+l]$  with lag  $l$ .

- **Gamma computation Unit ( $\Gamma U$ ):**  $\Gamma U$  consists of the Gamma computation sub-units  $\Gamma U_m$  that compute timing error functions  $\Gamma_m$ ,  $1 \leq m \leq M-1$ . From (4) and (7), sub-Unit  $\Gamma U_m$  is illustrated in Fig. 7 for  $1 \leq m \leq M-2$ . From (4) and (10), the sub-Unit  $\Gamma U_{M-1}$  is shown in Fig. 8. Sub-Unit  $\Gamma U_0$  is not shown herein because the timing error function  $\Gamma_0$  is not used for the timing skew estimation, see (25). However, the architecture of  $\Gamma U_0$  can be easily derived based on (4) and (10).
- **Matrix Processing Unit (MPU):** The MPU performs the matrix product in (27) to estimate the clock skews of sub-ADCs. Given  $M$ , constant elements of the pseudo-inverse constant circulant matrix  $\mathbf{A}^\dagger$  are computed in advance, stored in memory buffers and referred to as input parameters of the proposed calibration algorithm. As detailed

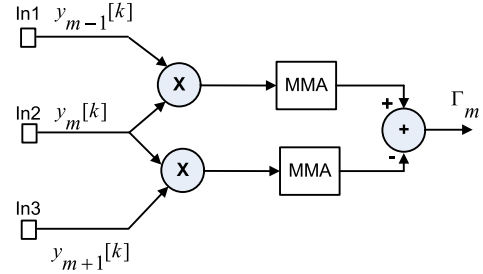
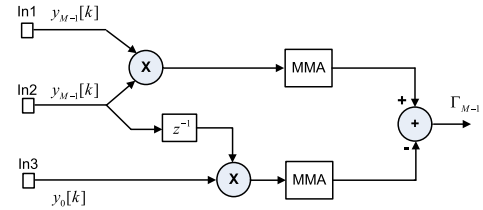
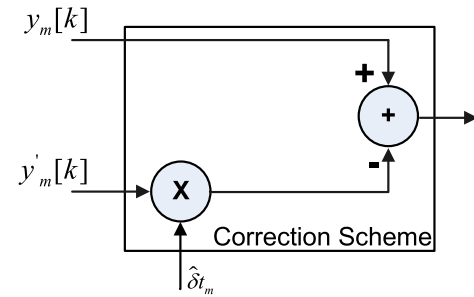
Fig. 7. Sub-unit  $\Gamma U_m$ ,  $1 \leq m \leq M-2$ .Fig. 8. Sub-unit  $\Gamma U_{M-1}$ .

Fig. 9. Correction scheme.

in Section III-B, timing error function components of the vector  $\Gamma$  are calculated based on cross-correlation being the average of  $N$  cross-products of sub-ADC samples. The aforementioned MMA filter is used to compute this average, i.e., computing the mean power. Moreover, the determination of the mean power comprises the use of period  $N$  samples containing the current sample and a plurality of preceding samples of the corresponding sequence. Thus, MMA filter enables frequent updating of the estimate value of the mean power, leading to a temporal track of the variations of the timing skew estimates due to Process-Voltage-Temperature (PVT) variations, aging or any kind of variations.

- **Correction Scheme:** The corrected samples  $\hat{y}_m[k]$  of the  $m^{th}$  sub-ADC are computed by first-order Taylor approximation according to (32). This principle is shown in Fig. 9.

$$\hat{y}_m[k] \approx y_m[k] - \hat{\delta}_{t_m} y'_m[k]. \quad (32)$$

#### D. Calibration for Input at Any Nyquist Band

As elaborated in Sections III-B, III-A and III-C, in order to cancel out the timing skew error, the derivative of the

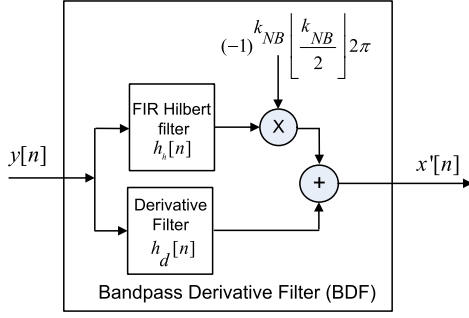


Fig. 10. Bandpass derivative filter.

WSS input bandlimited to the Nyquist frequency is determined by using the polyphase filter decomposition of the ideal differentiator filter with impulse response  $h_d[n]$ . Let us consider a continuous time WSS Bandpass (BP) input signal inside the  $k_{NB}^{th}$  NB. Its frequency content in the two frequency bands is then defined by

$$(k_{NB} - 1) \frac{f_s}{2} < f_L \leq |f| \leq f_H < k_{NB} \frac{f_s}{2}, k_{NB} \geq 1, \quad (3)$$

where  $f_L, f_H$  are the low and high cutoff frequencies of the input, respectively. This BP input occupies the  $k_{NB}^{th}$  Nyquist zone. If the condition (33) is fulfilled, there will be no aliases after sub-sampling the original input [37]. In [19], [27], a filter is proposed to compute the derivative of the original BP input of the undersampling (or sub-sampling) TIADC. The filter is called a BP Derivative (BD) filter and re-sketched in Fig. 10. It encompasses a scaling factor dependent on the order of NB  $k_{NB}$  and two Finite Impulse Response (FIR) filters with constant coefficients: a differentiator  $h_d[n]$  and a Hilbert filter. The Hilbert filter is an all-pass filter that shifts the input signal phase by 90 degrees [37] and its impulse response  $h_h[n]$  is expressed by [30], [37]

$$h_h[n] = \begin{cases} \frac{2}{\pi} \frac{\sin^2(\frac{\pi n}{2})}{n} & (n \neq 0) \\ 0 & (n = 0) \end{cases} \quad (34)$$

The impulse response of the BD filter is expressed by

$$h_{bd}[n] = h_d[n] + h_h[n] \times (-1)^{k_{NB}} \left[ \frac{k_{NB}}{2} \right] 2\pi. \quad (35)$$

The constant (or scale factor) of  $(-1)^{k_{NB}} \times \left[ \frac{k_{NB}}{2} \right] \times 2\pi$  is referred to as an input parameter of the proposed calibration algorithm. Given the order of NB, the BD filter has constant coefficients. By decomposing the  $M$ -component polyphase filter structure for the BD filter  $h_{bd}[n]$  as done for the aforementioned differentiator  $h_d[n]$ , the overall architecture of the proposed calibration illustrated in Fig. 3 is applicable for input at any NBs.

#### IV. EXPERIMENTAL RESULTS

##### A. Simulation Results

To verify the efficiency of the proposed technique, simulations are carried out on an undersampling four-channel TIADCs with 60dB SNR (thermal noise level) clocked

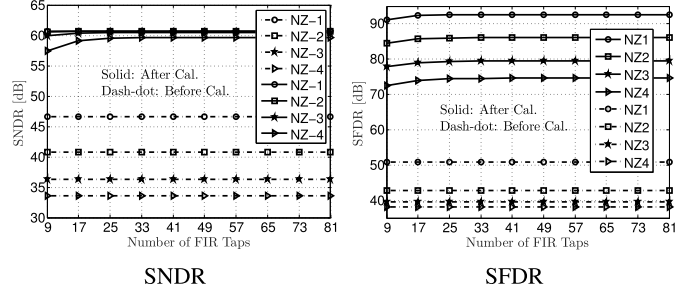


Fig. 11. SNDR/SFDR performance vs. number of FIR taps.

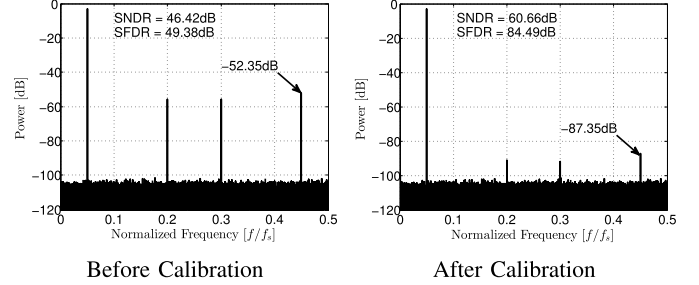


Fig. 12. Output spectrum of four channel TIADCs with input  $f_{in} = 0.45 \times f_s + \frac{f_s}{2}$  in second Nyquist band (due to undersampling TIADC,  $f_{in}$  maps to  $0.05 \times f_s$  in baseband).

at  $f_s = 2.7\text{GHz}$ . Timing skews are modeled as Gaussian distribution with zero mean and standard deviation  $\delta_{te}$  of 0.33ps. The constant smoothing factor  $\alpha$  of MMA filters is chosen as  $2^{-15}$  to achieve a good compromise between the convergence speed and the parameter estimation precision [38]. The NB order,  $k_{NB}$  of the input can be up to the fourth Nyquist band for the next-generation sub-sampling radio receivers in order to choose a proper and feasible sampling rate [29], [39]. Thus, the simulations demonstrating the efficiency of the proposed calibration are therefore performed up to the fourth NZ i.e., simulations with  $k_{NB} = \{1, 2, 3, 4\}$ .

The number of FIR taps is designed to be equal for both derivative filter and Hilbert filter. The coefficients of these FIR filters are obtained by multiplying the exact coefficients by the Hanning window to mitigate the influence of a truncation error. Their polyphase filters are causal and linear phase FIR filters. The coefficients of polyphase filters are explicitly computed based on equations (28), (29), (34) and (35). Note that the constant group delay of the polyphase filter of  $\frac{\text{number of FIR taps}-1}{2M}$  is added to all signal paths to balance the delay. In order to avoid the non integer delay, the number of FIR taps of the derivative and Hilbert filters should be an integer multiple of  $2M$  plus 1 [19]. The SNDR and SFDR performance versus the FIR taps over the first four NBs is drawn in Fig. 11. The solid and dash-dot curves show the performance with/without calibration, respectively. The curves with  $\circ, \square, \star$  and  $\triangleright$  markers show the SNDR/SFDR performance in respect of the first four NZs, respectively. As can be noticed, the optimal number of FIR taps is 25 taps at which the performance saturates.

Fig. 12 shows the output spectrum of TIADC before and after calibration for a single-tone sinusoidal input signal at

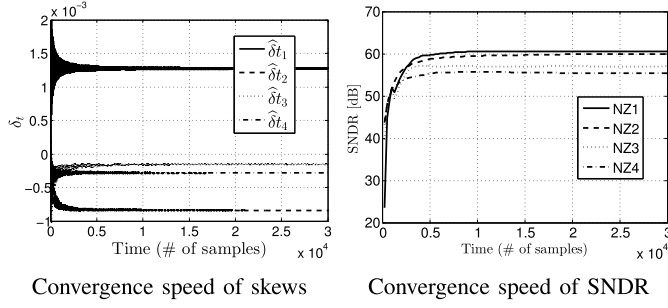


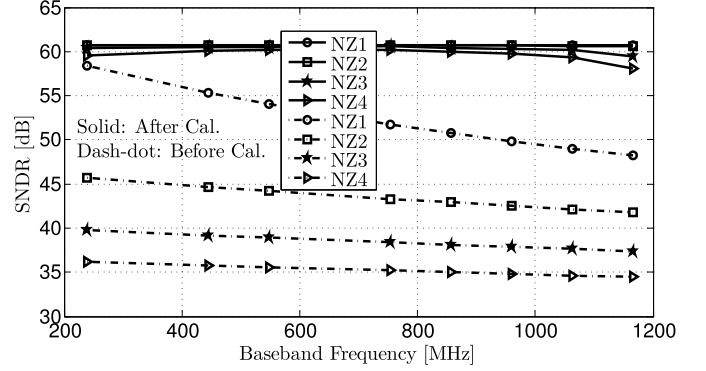
Fig. 13. Convergence speed behavior during calibration.

TABLE I  
COMPARISON WITH THE STATE-OF-THE-ART TECHNIQUES

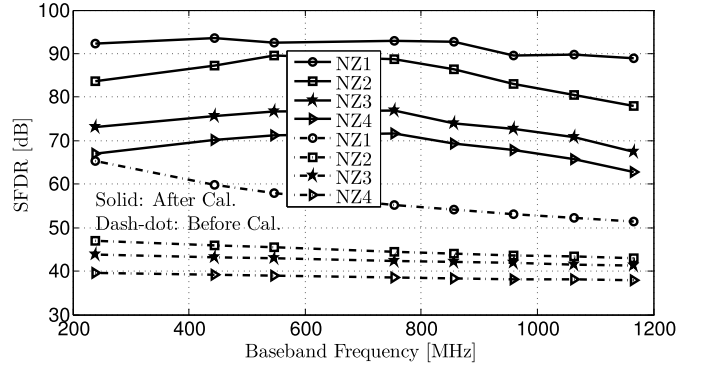
| Characteristics              | ISSCC<br>2014 [10] | TCAS-I<br>2012 [21] | SPAWC<br>2011 [22] | TCAS-II<br>2015 [19] | TCAS-I<br>This work |
|------------------------------|--------------------|---------------------|--------------------|----------------------|---------------------|
| Background                   | Yes                | Yes                 | Yes                | Yes                  | Yes                 |
| Blind                        | Yes                | Yes                 | semi-blind         | Yes                  | Yes                 |
| Input in any NB              | No                 | Yes                 | Yes                | Yes                  | Yes                 |
| Add ref. channel             | No                 | Yes                 | No                 | No                   | No                  |
| Pilot input injection        | No                 | No                  | Yes                | No                   | No                  |
| $M$ (# of Channels)          | 12                 | 4                   | 2                  | 4                    | 4                   |
| Clock freq. of filter        | $f_s$              | $f_s$               | $f_s$              | $f_s/M$              | $f_s/M$             |
| Cal. Manner                  | FF                 | FB                  | FB                 | FB                   | FF                  |
| Conv. time<br>[# of samples] | 10K                | 1.5K                | 4K                 | 10K                  | 5K                  |

the frequency of  $f_{in} = 0.45 \times f_s + \frac{f_s}{2}$  in the second NB. As illustrated in Fig. 12(b), the spurs due to the timing skews are mitigated significantly in comparison with the output spectrum before calibration shown in Fig. 12(a). The SFDR is improved by almost 35dB. The SNDR value after calibration is 60dB which is equal to its value in the no-mismatch case. Fig. 13(a) illustrates the convergence speed of timing mismatch estimates. The Monte Carlo simulation method is used to generate many iterations of clock skews in order to show the convergence speed of SNDR during calibration for the single-tone sinusoid input at a frequency of  $0.45 \times f_s + (k_{NB} - 1) \times f_s, k_{NB} \in \{1, 2, 3, 4\}$ . The SNDR curves as functions of time are shown in Fig. 13(b). By comparing Fig. 13(a) and Fig. 13(b), it can be seen that the clock skew estimates converge to their expected values after 5-K samples (or after  $1.8\mu s$ ). Note that the convergence time depends on the simulated parameters such as input frequency, number of channels, and channel mismatches. To analyze the key differences between the proposed calibration and the state-of-the-art techniques, Table I presents the reported available convergence time and main characteristics of prior arts.

Fig. 14 illustrates SNDR and SFDR versus the baseband frequencies to which the subsampling TIADC folds the input frequency at higher NB back. As can be seen, the SNDR and SFDR before calibration decrease with the input frequency and the NZ order increment. This is because the impact of timing skew increases with input frequency [3]. After calibration, the SFDR remains smaller for higher NZs. This is due to the fact that (i) (8) and (2) are obtained by employing the Taylor series to the first derivative assuming the sample-time error is small; (ii) the derivative signals in (2) and (19) are computed using the uncorrected sub-ADC outputs instead of input. These approximations become less accurate when the input frequency increases because the distortion level rises. Nevertheless, the



(a) SNDR



(b) SFDR

Fig. 14. Minimum SNDR/SFDR vs. input frequencies over the first four NBs: (a) SNDR; (b) SFDR.

proposed technique achieves an SFDR improvement of at least 28dB and 60dB SNDR over the first four NBs, which proves the efficiency and added value of the proposed technique.

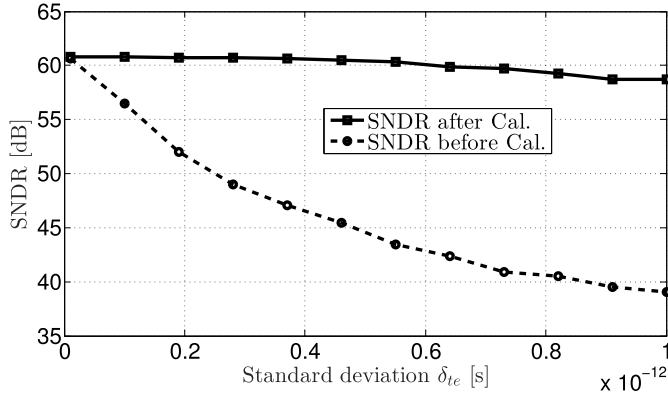
Fig. 15 shows the SNDR and SFDR versus the standard deviation  $\delta_{te}$  of clock skews for the input at frequency  $f_{in} = 0.45f_s + \frac{f_s}{2}$  in the second NZ. The solid and dash-dot curves show the performance after and before calibration, respectively. SNDR and SFDR before calibration decrease when timing skews increases. This is because the impact of timing mismatch rises with clock skews. The proposed calibration significantly improves the linearity up to 97dB SFDR for  $\delta_{te}$  less than 0.2ps.

The presented feedforward calibration is also validated for a band-limited bandpass multitone input signal. Fig. 16 shows the output spectrum with/without calibration for a 47 sinusoidal tone input in the second NB with  $f_L = 0.05f_s + \frac{f_s}{2}$ , and  $f_H = 0.4f_s + \frac{f_s}{2}$ . As illustrated in Fig. 16, undersampling TIADC directly down-converts the bandpass input to the frequency baseband  $(0, \frac{f_s}{2})$ , i.e.,  $f_H$  and  $f_L$  map to  $0.1f_s$  and  $0.45f_s$  due to the second NZ sub-sampling, respectively. It can be seen that spurs due to timing skews are reduced to the noise floor. The proposed technique obtains an SNDR improvement of approximately 14dB.

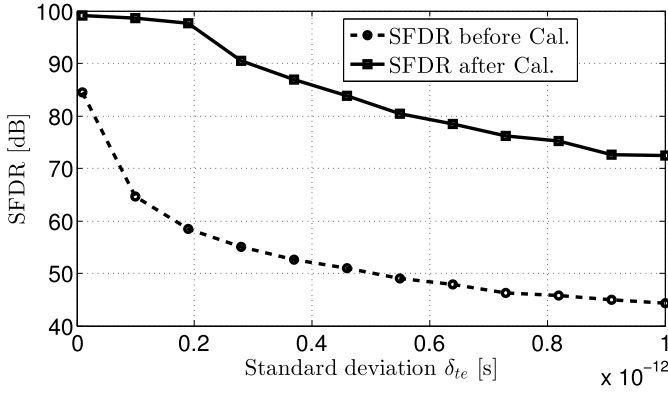
### B. Hardware Implementation and Validation

FPGA and ASIC design flow using MATLAB/Simulink in [19], [24], [26] is applied in this framework. The hardware

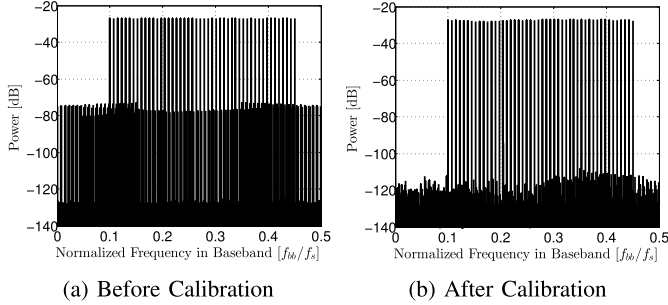




(a) SNDR



(b) SFDR

Fig. 15. SNDR and SFDR vs. the standard deviation  $\delta_{te}$  of clock skews for the input at frequency  $f_{in} = 0.45f_s + \frac{f_s}{2}$  in the second NZ.Fig. 16. Output spectra for the band-limited bandpass multitone input  $x(t)$  with  $f_L = 0.05f_s + \frac{f_s}{2}$ ,  $f_H = 0.4f_s + \frac{f_s}{2}$  in the second NZ (a) before calibration: SNDR = 43.9dB and (b) after calibration: SNDR = 57.4 dB. (Cutoff frequencies of  $f_H, f_L$  map to  $0.1f_s$  and  $0.45f_s$  in baseband, respectively).

architecture of the proposed calibration is designed and optimized in terms of fixed-point representation of signals that is characterized by signal ranges and signal *Word-Length* (WL). So the parameters of the Optimal Fixed-point Simulink (OFpS) model that need to be optimized, are the order of FIR filters and the signal WLs. The signal ranges of block signals in the OFpS model are mathematically computed based on the transfer function of DSP blocks and simulations as done in [24]. The signal ranges and signal WLs would determine the fractional bits (or precision) used to convert signal values into a binary representation. The signal

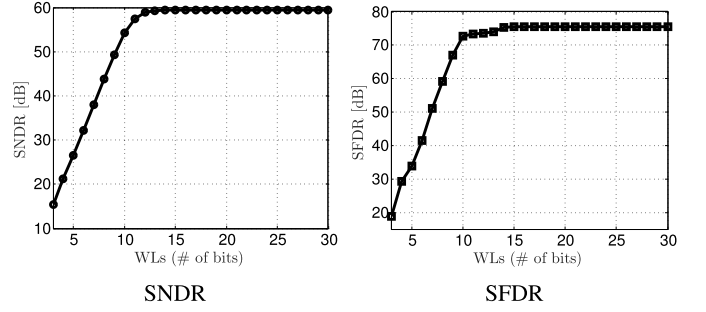


Fig. 17. Minimum SNDR/SFDR vs. WLs of the corrected sub-ADC outputs: (a) SNDR; (b) SFDR.

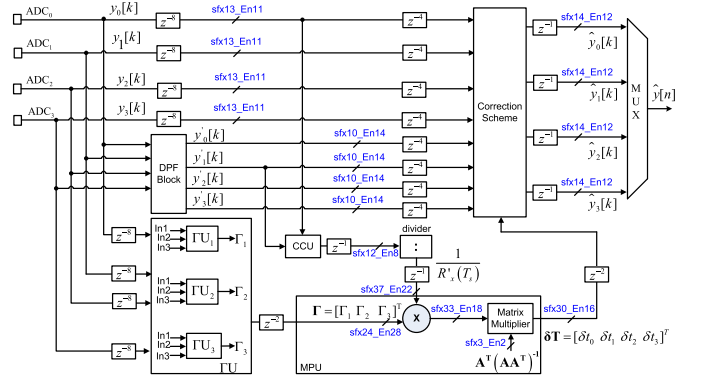


Fig. 18. The OFpS architecture of the proposed calibration for four-channel TIADCs.

WLs and the FIR orders impact SNDR/SFDR performance. Thus, they are optimized based on SNDR/SFDR metrics as presented in Fig. 11 for the FIR order optimization, and in Fig. 17 for the optimal WL of the corrected sub-ADC outputs. As can be seen in Fig. 17, the number of bits assigned to the compensated sub-ADC outputs is 14 bits.

The OFpS model (or hardware architecture) processes real-time signal data in sample-by-sample manner. Therefore, the delays of each signal datapath are made balanced. Note that, pipeline registers are inserted to reduce the combinational path length and improve the global working frequency and throughput [19]. Fig. 18 shows the pipelined OFpS architecture of the proposed calibration.  $z^{-k}$  blocks are delayed/pipeline registers. The notation of sfix13\_En11 presents a 13-bit signed fixed-point data type with a Fraction Length (FL) or fractional bits of 11, i.e., WL = 13 bits and FL = 11 bits. Generally, the representable fixed-point number is actually bounded by  $(-2^{WL-1-FL}, (2^{WL-1} - 1)2^{-FL})$ . The sub-ADC outputs are represented on 13 bits to decrease the sensitivity to digital truncation errors. System-level simulations reveal that the corrected sub-ADC outputs are presented by 14 bits for minimum digital truncation errors. Note that since the delayed/pipeline registers are manually inserted at the outputs and inputs of polyphase filters subPF<sub>i</sub>, the total number of delayed/pipeline registers of the DPF is 8. As shown in Fig. 18, the latency between the distorted output and the compensated output is 13 clock cycles.

The OFpS model is converted into Hardware Description Language (HDL) using MATLAB HDL Coder toolbox.

In this framework, Verilog code is generated automatically for the FPGA/ASIC design target. The Hardware-In-Loop (HIL) emulation methodology in [19], [24], [26] is applied to validate the proposed calibration on an Altera FPGA board. Simulation shows that the synthesized circuit operates properly on the FPGA and dissipates very little hardware resources of the FPGA chip.

1) *ASIC Synthesis*: The ASIC design flow proposed in [19], [26] is applied in this framework. The HDL design is synthesized to a gate-level netlist using a Cadence RTL Compiler (RC) targeting the ST-28nm FD-SOI technology. The RC synthesis tool uses the automatic datapath retiming techniques in order to improve the working frequency. To make more efficient power savings, the low power (RC-LP) engine automatically inserts the clock-gating logic for register banks. The RC-LP engine identifies when the registers are inactive and disables the clock during these periods. Once timing requirements are fulfilled, automatic placement and routing are performed using a Cadence Encounter Place and Route (P&R) tool. During the P&R phase, a clock tree synthesis is also performed and the clock buffers are placed to ensure correct clock propagation and synchronization in the design [19]. The P&R logic simulations are performed using the P&R Verilog gate-level netlist and the Verilog testbench generated by MATLAB HDL coder Toolbox. The Verilog Value-Change Dump (VCD) file generated by P&R logic simulations contains the information about value changes on selected signals. Reading the switching activity information from a VCD file by the Encounter RTL compiler provides the detailed information about the switching behavior of nets and ports, which leads to more accurate power estimation [19].

Note that the optimization of the fixed point coding, the filter coefficients and the signal path Word-Length (WL), is performed using the MATLAB-Simulink. Thanks to this optimization methodology, the Verilog code (or RTL design) was created automatically from the above designed OFPs model. The output of RTL simulations in Modelsim is identical to the output of MATLAB Simulink simulations. Thus, if there is no error after the post P&R logic simulations, the achieved output of the synthesized circuit after the layout simulation is the same as the output of the fixed-point Simulink model.

Using the polyphase implementation, the working clock frequency of the synthesized circuit is  $\frac{f_s}{4} = 675\text{MHz}$ . The post P&R logic simulations are executed without any error messages at 2.7GHz, i.e., reporting that Total Negative Slack (TNS = 0), Worst Negative Slack (WNS = 0.78ns), and there are no violating paths. The overall speed of the calibration system is 2.7GHz. It consumes a total power of 33.2mW at an input frequency of 5.3GHz in the fourth NZ and occupies an area of 0.04mm<sup>2</sup>. The synthesized calibration system uses 26032 logic gates in total as shown in Table II.

Fig. 19(a) shows the pie chart of the power consumption at the level hierarchy of the synthesized system. As can be seen, the DPF block dissipates 45% of the total power, significantly more power than the other sub-modules. Fig. 19(b) displays the physical locality of the sub-modules on the chip of the synthesized and designed digital circuit. The DPF block

TABLE II  
THE TARGETED LOGIC GATES FOR SUB-SAMPLING ADCs

| Type       | Instances | Area [ $\mu\text{m}^2$ ] | Area % |
|------------|-----------|--------------------------|--------|
| Sequential | 4185      | 18367.181                | 41.6   |
| Inverter   | 5905      | 1963.459                 | 4.4    |
| Buffer     | 449       | 240.067                  | 0.5    |
| Logic      | 15493     | 23600.106                | 53.4   |
| Total      | 26032     | 44170.813                | 100.0  |

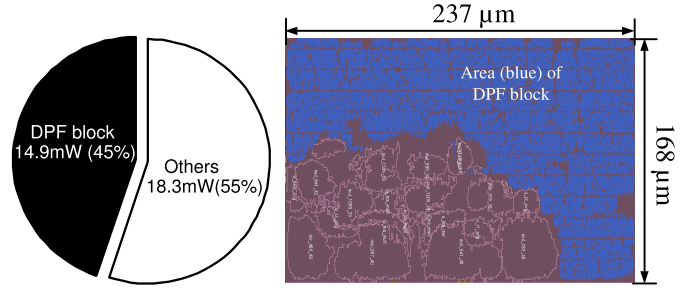


Fig. 19. (a) Power pie chart and (b) module locality of the chip die area.

TABLE III  
THE TARGETED LOGIC GATES FOR BASEBAND ADCs

| Type       | Instances | Area [ $\mu\text{m}^2$ ] | Area % |
|------------|-----------|--------------------------|--------|
| Sequential | 2255      | 9864.787                 | 47.1   |
| Inverter   | 992       | 338.477                  | 1.6    |
| Buffer     | 55        | 34.272                   | 0.2    |
| Logic      | 7912      | 10716.528                | 51.1   |
| Total      | 11214     | 20954.064                | 100.0  |

formed from the digital polyphase filter components of the BD FIR filter occupies the biggest part of the chip die area.

The above calibration system design works properly for the sub-sampling ADC with the input up to NZ4. Considering the baseband (or Nyquist) ADC applications, the calibration is designed for the input in the first NB. In this case, the digital background calibration unit dissipates 15.5mW total power at a input frequency of 1.2GHz in the first NZ and occupies 0.02mm<sup>2</sup> of the chip area. As presented in Table III and in comparison with Table II, the number of logic gates and area of the calibration hardware architecture designed for baseband ADCs, are reduced greatly compared to the sub-sampling ADCs. Moreover, its power dissipation is 50% lower than that of the calibration hardware architecture dedicated for sub-sampling ADCs. This is because the calibration hardware for sub-sampling ADCs with the input up to the NZ4 requires higher dynamic ranges of all block signals and higher WLs compared to that for baseband ADCs. Moreover, in addition to the Nyquist derivative filter, we have a FIR Hilbert filter as well as the NZ order dependent scale factor as shown in Fig. 10. In practice, the choice of the calibration architecture will depend on applications. If it is a sub-sampling application, then the design presented in Table II will be used and in classical ADCs, the design of Table III will be used in order to minimize power consumption and area.

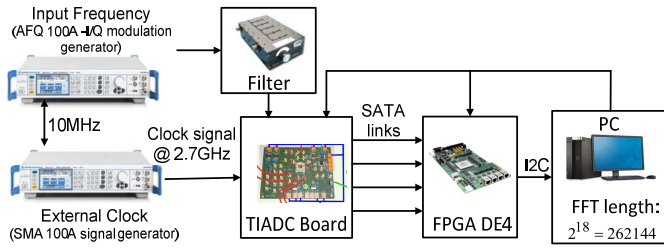


Fig. 20. Lab environment and setup measurement.

2) *Measurement Set-Up*: This section is dedicated to the results achieved by post processing the measured data using the TIADC board provided by NXP Semiconductor and published in [6]. The chip is a 64-channel 11-b time-interleaved Successive-Approximation-Register (SAR) ADC clocked at 2.7GHz. 64 channels are divided into 4 quarters. Each quarter consists of the 16 SAR ADCs which are time-interleaved in parallel and share the same sample and hold. The effect of the timing mismatch in this TIADC is therefore equivalent to the timing mismatch in four-channel TIADCs. This is the reason why the above simulations are carried out for the four-channel TIADC in order to demonstrate the efficiency of our solution. Note that offset and gain mismatches are static and frequency independent. Their impacts are the same for both subsampling and regular baseband TIADCs [19], [27]. Moreover, the offset and gain calibrations integrated on the ADC chip can be turned on/off by software. Thus, all-digital offset and gain mismatch calibrations are proposed in this framework in order to tackle all channel mismatches in subsampling TIADCs with the real samples captured from the chip.

A simple way of canceling the offset error is to calculate the modified moving average of each sub-ADC output and then subtract the average results from their respective sub-ADC output samples [27]. The offset-corrected samples are then passed into the gain mismatch calibration unit where gain mismatch is mitigated. The relative gain of each sub-ADC with respect to a reference sub-ADC is determined by computing a power ratio between the average powers of sub-ADCs and the reference channel. The relative gain estimate is multiplied by the according sub-ADC output samples to generate the corrected sub-ADC output [26]. The gain-corrected channels have the same gain values as the reference channel, hence equalizing gain mismatch among ADC channels.

The offset and gain corrected samples are then transmitted to the pipelined OFpS architecture where clock skews are corrected by our proposed calibration. The pipelined OFpS model is in fact equivalent to the synthesized digital circuit as interpreted in Section IV-B1. Thus, the emulated results also present the efficiency of our solution when it is integrated on the ADC chip.

The measurement set-up used to measure the TIADC output, is described in Fig. 20. The equipment is set up as follows:

- Arbitrary signal generators up to 6GHz.
- A 2.7GHz external clock generator needs to supply the clock signal to the ADC chip. The input signal generators and the clock signal generator are synchronized through a reference signal (10MHz).

- Narrow band filters are used for the input signal frequency to remove the harmonics (or distortions) created by the input signal generators and achieve a pure sine-wave at the input of the ADC.
- Because of the limited Random-Access Memory (RAM) (limited to 16K samples) of the ADC chip, the TIADC is connected to a FPGA DE4 board via a SATA connection. The SATA interface is made of the two differential links connected to the FPGA board. Because of the limited speed of the SATA bus, the ADC output is downsampled by 5. The downsampled output sequence is saved in the FPGA RAM that can store up to  $2^{18} = 262144$  samples. This determines the resolution of the power spectral density (PSD).
- Measured data is transferred to a Personal Computer (PC) via an I2C connection for post processing calibrations.

For the measurement, the offset and gain calibrations embedded on the chip are turned off. The ADC full scale is 1.2V. A two-tone sinusoidal input is created at the frequencies of  $f_{in} = [2300\text{MHz}, 2301\text{MHz}]$  in the second NB. The two tones have the same power of  $-14\text{dBFS}$ . The frequency gap between the two tones is 1MHz. This is because the generated input is guaranteed inside the bandwidth of a narrow band filter. Note that the ADC chip was designed and optimized to work in the first NZ. However, to test our calibration, we used it in the second NZ where it has some attenuation. At 2.3GHz, the measured attenuation is around 6dB, that is why the fundamental tones appear at the power of  $-20\text{dBFS}$  at the output spectrum as shown in Fig. 21.

Fig. 21(a) and (b) show the output spectrum before and after executing the proposed all-digital offset and gain calibrations, respectively. The output of the ADC chip is distorted by channel mismatches (offset, gain and clock skew) and non-linear harmonic distortion coming from the electrical components of the ADC front-end. Note that due to sub-sampling and downsampling processes, input tones map to 140MHz and 141MHz, respectively as shown in Fig. 21. Obviously, the offset tones are completely removed after calibration. The spurs due to timing skew and gain mismatches have the same position at the output spectrum [3]. As can be seen, the gain calibration compensates for the real samples and reduces the level of some (but not all) spurious tones induced by clock skews and gain mismatches by around 3dB. The corrected output samples after the offset and gain calibration are fed into the optimal hardware architecture illustrated in Fig. 18 to mitigate the clock skew errors. Fig. 21(c) illustrates the output spectrum after performing the proposed calibration algorithm. As can be seen, the proposed clock skew calibration significantly mitigates the skew and gain spur levels to the noise floor level from  $-66\text{dBFS}$  to  $-97\text{dBFS}$ . During calibration, the timing mismatch coefficients converge to their expected values after 5K samples (or  $1.8\mu\text{s}$ ) as shown in Fig. 22.

From the above measurement results, the ADC chip suffers from nonlinear distortions coming from the front-end of the ADC. The effects of these distortion errors can be removed by applying the digital distortion compensation techniques presented in [40], [41].



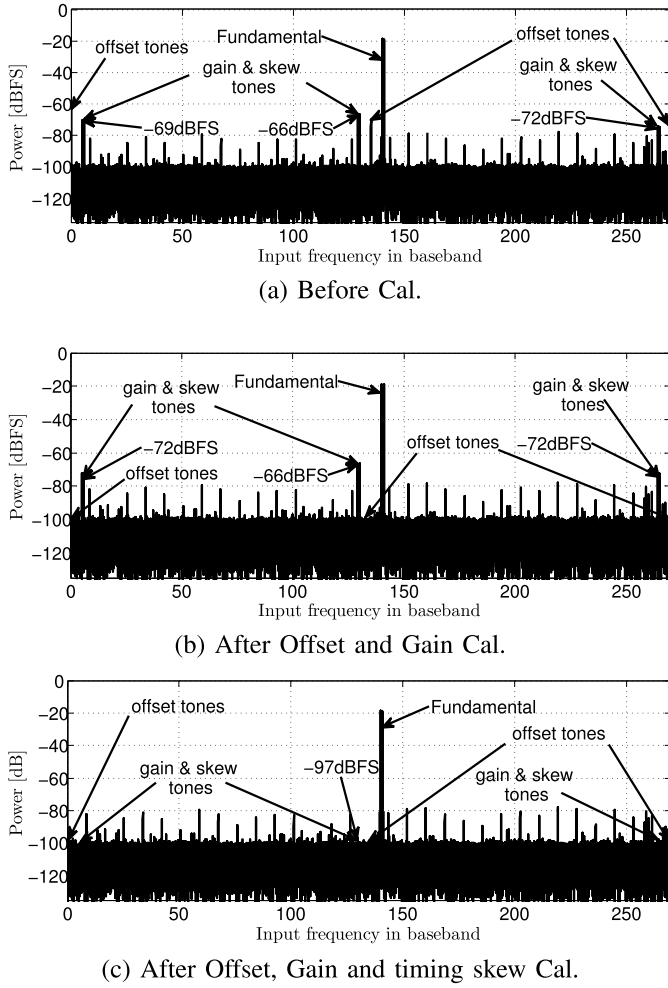


Fig. 21. Measured output spectrum after and before calibration (due to sub-sampling and downsampling two original input tones map to 140MHz and 141MHz, respectively).

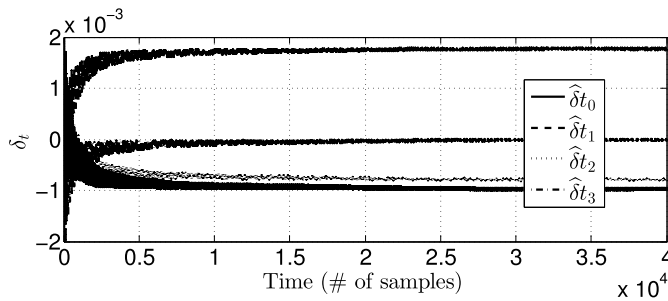


Fig. 22. Convergence speed of clock skews with post-processing the real samples.

By making a survey of the TIADC chips published at the ISSCC and VLSI conferences from 1997 to 2016 [42], if the clock skew calibration is needed, most TIADC chips used the mixed-signal calibration. Table IV shows the performance comparison of the synthesized digital logic of only clock skew calibrations to the state-of-the-art techniques. The designed digital logic of the proposed calibration system successfully tackles the timing skew problem for input at any NB. In post-processing simulations with the real data captured

TABLE IV  
PERFORMANCE COMPARISON

| Ref.         | This Work       | Our previous work [26] | ISSCC 2002 [43] | ISSCC 2014 [10] |
|--------------|-----------------|------------------------|-----------------|-----------------|
| Technology   | 28nm FD-SOI     | 28nm FD-SOI            | 0.35- $\mu$ m   | 40nm            |
| M            | 4               | 4                      | 2               | 12              |
| Rate [GS/s]  | 2.7             | 2.7                    | 0.12            | 1.6             |
| Resolution   | 11 bits         | 11 bits                | 10 bits         | 12 bits         |
| Input        | Up to NZ4       | Up to NZ4              | NZ1             | NZ1             |
| Spurs [dBFS] | 97 @ $f_{in}^1$ | 97 @ $f_{in}^1$        | 90.3 @ 0.99MHz  | 70 @ 750MHz     |
| Power [mW]   | 33.2            | 41                     | 171             | 35.3            |

<sup>1</sup>  $f_{in} = \{2300\text{MHz}, 2301\text{MHz}\}$

from the ADC chip, it keeps skew tones at -97dBFS for two-tone sinusoidal input located in the second NZ at frequencies of 2300MHz, 2301MHz. Moreover, it dissipates less power and works at higher sampling rate than the prior arts. As clock skew induced spur levels are reduced by 31dB, the proposed calibration has the same emulation performance as our previous work reported in [19], [26]. However, it consumes 7.5mW less power after post P&R logic simulations, reducing the power consumption by 19%. This is due to the use of only one BD filter instead of the two filters used in [19]. In addition, considering the baseband ADC clocked 2.7GHz and as presented in Section IV-B1, the synthesized calibration system consumes 15.5mW. This proves that our solution outperforms state-of-the-art calibrations in terms of low power consumption and high sampling rate.

## V. CONCLUSION

This framework has presented an all-digital clock skew feedforward background calibration for sub-sampling TIADCs. This technique does not require a pilot input nor an additional reference channel. It is implemented using the polyphase filtering technique in order to enhance the working frequency in the DSP. It also does not use adaptive filter banks. Thus, this solution requires less hardware and consumes less power. Moreover, the correction and estimation algorithms in the feedforward calibration scheme re-use (or share) the common BD FIR filter which allows to save filter hardware cost and to reduce power dissipation by 19%. Simulations demonstrate the efficiency of the proposed calibration which leads to an SFDR improvement of at least 28dB over the first four NZs. The pipelined hardware architecture of the proposed calibration is optimized using the fixed-point optimization methodology. The HDL design of the pipelined hardware architecture has been synthesized using the RC targeting ST-28nm FD-SOI process. The designed digital circuit successfully addresses the clock skew problem for the 60dB SNR sub-sampling and baseband TIADC applications clocked at 2.7GHz. For the undersampling ADC application, it occupies an area of 0.04mm<sup>2</sup> and dissipates a total power of 33.2mW at the input frequency of 5.3GHz. For the baseband ADC application, it occupies 0.02mm<sup>2</sup> and dissipates 15.5mW at the input frequency of 1.2GHz. In comparison with the state-of-the-art performances, our solution achieves lower power consumption and higher rate with a small chip area. As post-processing the measured data from the ADC chip, the proposed clock skew calibration incorporates the classical calibration algorithms of



gain and offset mismatches in order to successfully address all channel mismatches. The clock skew calibration keeps the timing mismatch tones below  $-97\text{dBFS}$  for the two-tone sinusoidal input in the second Nyquist Zone after processing the measured data stream from the ADC chip, demonstrating the outperformance of our solution compared with prior arts.

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